

ROB STEILBERG

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EDUCATION

Duke University, B.S. Computer Science

2014 – 2018

EXPERIENCE

Jeppesen ForeFlight

Austin, TX

Senior Software Engineer II, 2026 – present

- Defined team-wide best practices on LLM usage; established standards for AI-first development and code review
- Contributed to the buildout of ForeFlight's first MCP server, now available for use in ChatGPT and Claude
- Designed and built backend services that package, sign, verify, and publish Jeppesen navigation database updates for Garmin and Honeywell avionics

Senior Software Engineer I, 2022 – 2026

- Designed and built the backend for Reported Turbulence, a real-time pipeline of turbulence data sourced from the accelerometers in ForeFlight's Sentry ADS-B receivers
 - Reused the architecture to import live airport weather camera imagery from the FAA and NavCanada for use in the ForeFlight Mobile EFB iOS application
- Represented ForeFlight at EAA AirVenture Oshkosh, SUN 'n FUN, and several aircraft owner conventions, providing in-person support, product expertise, and sales guidance

Server Developer, 2021 – 2022

- Shipped a complete frontend redesign (Laravel to React) and backend rewrite (PHP to Spring) of ForeFlight's retail and contract aviation fuel pricing features with minimal user disruption
- Triageed complex, user-specific issues related to contract fuel pricing, typically deploying fixes within a day
- Mentored an intern who was later hired as a full-time employee

Esri

Washington, DC

Software Engineer, 2018 – 2021

- Worked across the entire stack with the Esri R&D Center, contributing to UI/UX planning and feedback, frontend work with Ember.js, and backend work with Express.js, Elasticsearch, and PostgreSQL
- Rebuilt ArcGIS Hub's obsolete domain resolution service as a scalable microservice with Express.js, which enabled the shutdown of a legacy, inefficient Ruby on Rails app that reduced annual server costs by approximately \$100k, all with zero downtime
- Contributed to Esri's open source, cross-team TypeScript libraries, plus several popular Ember.js libraries
- Provided technical support and user education at the Esri User Conference and the Esri Developer Summit

Intern, summer 2017

- Prototyped an Ember.js app that templatizes ArcGIS Online mapping applications, making them compatible with any Esri-hosted dataset and improving app buildout time by over 85%

Clinica Esperanza

Roatán, Honduras | Spanish language proficiency

Volunteer, 2018 – present

- Directed UI/UX design, system architecture, and product development for mVax, an Android app used to record immunizations and generate standardized vaccination reports; led a team of six engineers
 - Completed onsite deployment in July 2018, increasing community vaccination uptake
- Continuing provider of periodic onsite IT support to clinic employees, in addition to ongoing website maintenance, strategy, and SEO optimization for nonprofit fundraising

TECHNICAL SKILLS

Current: Java (Spring Boot), JavaScript, Node.js (React, Express, Ember), HTML/CSS, Redis, Docker, nginx, SQL (PostgreSQL, SQLite, MySQL), AWS (CDK, CloudFormation, Route 53, ECS, S3, SNS, SQS, CloudFront, and others), avionics integration; familiar with Swift, Objective-C, TypeScript

AI / Developer Tooling: LLM-assisted development, prompt/workflow design, Claude/OpenAI/Codex-style tooling, MCP server prototyping

Previous: Python, Rust, C, C++, Ruby on Rails, R, Lisp (Racket), PHP, MIPS, Android, MongoDB, Firebase, jQuery, Elasticsearch, Kubernetes

PERSONAL WORK

roob1090

- A customized, dump1090-powered ADS-B data ingestion pipeline that drives a real-time flight tracking API, built with Rust, Python, Express.js, MongoDB, and Docker
 - A Raspberry Pi parses ADS-B signals decoded by dump1090 and then transmits aircraft position reports via WebSocket to an Express.js server
 - Individual flights are then geofenced into specific approach corridors at AUS and enriched with aircraft type and routing information from OpenSky and FlightAware

PATENTS

Aircraft Turbulence Notification System and Method US20250322753A1 (pending)

- Method for determining degree and impact of aircraft turbulence and then visualizing the result; filed through my work with ForeFlight on the Reported Turbulence feature

AVIATION EXPERIENCE

Certifications	Hours (August 2026)
Commercial Pilot, Instrument: ASEL, ASES, AMEL	Total: 1209.6
Certified Flight Instructor, Instrument: Airplane Single and Multiengine	PIC: 1071.8
Ground Instructor, Advanced and Instrument	Instrument: 126.5
FAA First Class Medical	Cross-country: 412.4
	Multiengine: 60.8
	Sea: 11.0
Types (specialties bolded)	Night: 132.1
C152/172/182, P28A/B/R, PA12, PA18, PA38, BE23/35/76, CH7B,	Complex/Retract: 167.4
COL3, AA1, BL8, FA62, PIVE	Tailwheel: 52.3

Self-Employed

Austin, TX

- FAA-certified provider of primary, instrument, commercial, multiengine, and recurrent flight instruction

Certified Flight Instructor, 2024 – present

Austin Air Cruisers

Austin, TX

- Flew a 1947 Piper PA-12 Super Cruiser on air tours over Austin and Lake Travis

Pilot, 2025 – 2026

INTERESTS

- SCUBA diving, holding PADI/SSI Open Water, Advanced Open Water, Rescue, Nitrox, and spearfishing diving certifications
- Global health and humanitarian work in Latin America, with additional experience in Guayaquil, Ecuador working at an equine therapy rehabilitation center for children with special needs